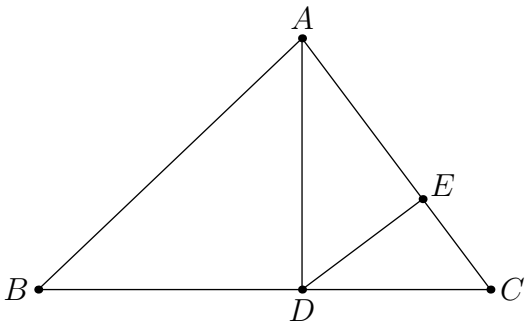
2024 SSMO Relay Round 1 Solutions

SMO Team

RR 1 Part 1: Let AD be an altitude of triangle ABC and let DE be an altitude of triangle ACD. If $AB = 29$, $CE = 9$, and $DE = 12$, find the area of triangle ABC.



Answer: $\boxed{360}$

Solution: From the Pythagorean Theorem, we have $DC = 15$. From similar triangles, $\frac{DC}{AD} = \frac{EC}{DE} = \frac{3}{4} \implies AD = 20$. So, $BD = \sqrt{29^2 - 20^2} = 21$. In conclusion,

$$[ABC] = \frac{1}{2}(BC)AD = \frac{1}{2}(BD + DC)(AD) = \frac{1}{2}(21 + 15)(20) = \boxed{360}.$$

RR 1 Part 2: Let $T = TNYWR$. A circular necklace is called *interesting* if it has T black beads and T white beads. A move consists of cutting out a segment of consecutive beads and reattaching it in reverse. Find the least positive integer x such that any *interesting* necklace can be changed into one in which all T black beads are consecutive using at most x moves.

Answer: $\boxed{359}$

Solution: Note that a move only changes the two pairs of adjacent beads at the ends of the reversed segment, since every other pair of adjacent beads stays adjacent. So, each move changes the number of places where a black bead is next to a white bead by at most 2, meaning it changes the number of blocks of consecutive black beads by at most 1. Since the necklace with alternating colors has T blocks of black beads and the target necklace has 1, at least $T - 1$ moves are needed. Conversely, if a necklace has at least 2 blocks of black beads, reversing a block of white beads together with the block of black beads after it merges two blocks of black beads, so $T - 1$ moves always suffice. Thus, $x = T - 1 = 360 - 1 = \boxed{359}$.

RR 1 Part 3: Let $T = TNYWR$. In a circle, there are T people. $T - 2$ of them have red shoes, and two of them have blue shoes. First, $\left\lfloor \frac{T}{2} \right\rfloor$ people with red shoes are eliminated one at a time, each chosen uniformly at random from the remaining people with red shoes. Then, people are eliminated one at a time, each chosen uniformly at random from everyone

remaining, until there is only one person left. The probability that this person has blue shoes can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{91}$

Solution: After the first phase, $T - \left\lfloor \frac{T}{2} \right\rfloor = 180$ people remain, and 2 of them have blue shoes. Since every elimination in the second phase is uniformly random, by symmetry each of the remaining people is equally likely to be the last one left. So, the probability is

$$\frac{2}{180} = \frac{1}{90} \implies 1 + 90 = \boxed{91}.$$

2024 SSMO Relay Round 2 Solutions

SMO Team

RR 2 Part 1: In a regular hexagon $ABCDEF$, let X be a point inside the hexagon such that $XA = XB = 3$. If the area of the hexagon is $6\sqrt{3}$, then XE^2 can be expressed as $a - b\sqrt{c}$, where a, b, and c are positive integers and c is not divisible by the square of any prime. Find $a + b + c$.

Answer: $\boxed{35}$

Solution: Let s be the side length of the hexagon. We have $\frac{6s^2\sqrt{3}}{4} = 6\sqrt{3} \implies s = 2$. Let h be the distance from X to AB. From the Pythagorean Theorem, we have $\sqrt{h^2 + \left(\frac{s}{2}\right)^2} = 3 \implies h^2 = 8 \implies h = 2\sqrt{2}$. Since the distance from AB to DE is $2\sqrt{3}$,

$$XE = \sqrt{(2\sqrt{3} - 2\sqrt{2})^2 + \left(\frac{s}{2}\right)^2} \implies XE^2 = 21 - 8\sqrt{6} \implies 21 + 8 + 6 = \boxed{35}.$$

RR 2 Part 2: Let $T = TNYWR$. If

$$a = \sum_{n=1}^{T} n(n+1)(n+2),$$

find the last three digits of a.

Answer: $\boxed{890}$

Solution: Note that

$$n(n+1)(n+2) = \frac{n(n+1)(n+2)(n+3) - (n-1)n(n+1)(n+2)}{4},$$

so the sum telescopes to

$$a = \frac{T(T+1)(T+2)(T+3)}{4} = \frac{35 \cdot 36 \cdot 37 \cdot 38}{4} = 442890,$$

meaning the last three digits of a are $\boxed{890}$.

RR 2 Part 3: Let $T = TNYWR$. Let $A = (0, 0)$, $B = (0, T)$, $C = (T, T)$, and $D = (T, 0)$. Find the perimeter of the set S of all points P inside square $ABCD$ for which $AP + CP \geq BP + DP$.

Answer: $\boxed{3560}$

Solution: Let $P = (x, y)$. Since both sides are nonnegative, we may square, cancel $x^2 + y^2 + (x-T)^2 + (y-T)^2$ from both sides, and square again. We have

$$\begin{aligned}
\sqrt{x^2+y^2} + \sqrt{(x-T)^2+(y-T)^2} \geq \sqrt{x^2+(y-T)^2} + \sqrt{(x-T)^2+y^2} &\implies \\
x^2+y^2+(x-T)^2+(y-T)^2+2\sqrt{x^2+y^2}\cdot\sqrt{(x-T)^2+(y-T)^2} \geq & \\
x^2+(y-T)^2+(x-T)^2+y^2+2\sqrt{x^2+(y-T)^2}\cdot\sqrt{(x-T)^2+y^2} &\implies \\
(x^2+y^2)((x-T)^2+(y-T)^2) \geq (x^2+(y-T)^2)(y^2+(x-T)^2) &\implies \\
y^2(x-T)^2+(x^2)(y-T)^2 \geq x^2y^2+(x-T)^2(y-T)^2 &\implies \\
(x^2-(x-T)^2)(y^2-(y-T)^2) \leq 0 &\implies \\
(T^2-2xT)(T^2-2yT) \leq 0 &\implies \\
\left(x-\frac{T}{2}\right)\left(y-\frac{T}{2}\right) \leq 0. &
\end{aligned}$$

So, if we split the square into four smaller squares by drawing lines $x = \frac{T}{2}$ and $y = \frac{T}{2}$, the set S consists of the top left and bottom right squares, each with side length $\frac{T}{2}$. Thus, the perimeter of the set S is $2 \cdot 4 \cdot \frac{T}{2} = 4T = 4 \cdot 890 = \boxed{3560}$.

2024 SSMO Relay Round 3 Solutions

SMO Team

RR 3 Part 1: Alice and Bob are on the second floor of a building, right next to the top of an escalator moving upwards. Alice and Bob each run at 200 steps per minute relative to the ground or the escalator they are on, and each escalator is a total of 225 steps. Alice chooses to run down the escalator moving upwards, while Bob chooses to run to the escalator moving downwards, 300 steps away, and then run down that escalator. Both escalators move at the same speed. If Alice and Bob hit the bottom floor at the exact same time, find the speed of the escalators in terms of steps per minute.

Answer: $\boxed{100}$

Solution: Let the speed of the escalators be s. We have the following Distance/Speed/Time chart.

	Distance	Speed	Time
Alice on the escalator	225	$200-s$	a_1
Bob running to the escalator	300	200	b_1
Bob on the escalator	225	$200+s$	b_2

Since Alice and Bob reach the bottom floor at the same time, we have $a_1 = b_1 + b_2$. Since distance is the product of speed and time, we have

$$\begin{aligned}
\frac{225}{200-s} &= \frac{3}{2} + \frac{225}{200+s} \implies \\
\frac{450s}{200^2-s^2} &= \frac{3}{2} \implies \\
s^2 + 300s - 200^2 &= 0 \implies \\
(s+400)(s-100) &= 0 \implies \\
s &= \boxed{100}.
\end{aligned}$$

RR 3 Part 2: Let $T = TNYWR$. Find the greatest odd integer n for which $n^2 + (T-1)n$ is a perfect square.

Answer: $\boxed{2401}$

Solution: Note that $T - 1 = 99$, so we have $n^2 + 99n = s^2$ for some nonnegative integer s. Factoring, we have

$$\begin{aligned}
\left(\left(n+\frac{99}{2}\right)+s\right)\left(\left(n+\frac{99}{2}\right)-s\right) &= \frac{99^2}{4} \implies \\
(2n+99-2s)(2n+99+2s) &= 99^2.
\end{aligned}$$

Since the sum of the two factors is $4n + 198$, n is maximized when the factors have the largest possible sum, which happens for the factor pair $1 \cdot 9801$. So, we let $2n + 99 - 2s = 1$ and $2n + 99 + 2s = 99^2 = 9801$. This gives $n = \frac{\frac{9801+1}{2} - 99}{2} = \boxed{2401}$, which is odd.

RR 3 Part 3: Let $T = TNYWR$. Riley and Boris are playing a game on a $(T-1) \times (T-1)$ grid of dots. The game alternates turns and starts with Riley. Each turn, a player draws a segment connecting two dots exactly 1 unit apart that are not already connected. The game ends when every such pair of dots is connected, and the player who draws the last segment loses. Given that Riley plays optimally and Boris plays randomly, the probability that Riley wins can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{2}$

Solution: Note that a $g \times g$ grid of dots has $g(g-1)$ horizontal and $g(g-1)$ vertical segments of length 1, so the game always lasts exactly $2g(g-1)$ turns, no matter how the players play. Since $2g(g-1)$ is even, Boris always draws the last segment on the $(T-1) \times (T-1) = 2400 \times 2400$ grid, giving a probability of $\frac{1}{1} \implies 1 + 1 = \boxed{2}$.

2024 SSMO Relay Round 4 Solutions

SMO Team

RR 4 Part 1: Freddy the Frog can jump 1 unit right or up. He is at $(0, 0)$ and wants to get to $(7, 4)$. However, he is scared of points $(3, 1)$ and $(3, 2)$ and will not hop onto those points. How many ways can he reach his destination?

Answer: $\boxed{100}$

Solution: Using the Principle of Inclusion and Exclusion, the answer is the number of ways to get to $(7, 4)$ minus the number of ways to get to $(7, 4)$ passing through $(3, 1)$ minus the number of ways to get to $(7, 4)$ passing through $(3, 2)$ plus the number of ways to get to $(7, 4)$ passing through both $(3, 1)$ and $(3, 2)$. This is equivalent to

$$\binom{11}{4} - \binom{4}{1}\binom{7}{3} - \binom{5}{2}\binom{6}{2} + \binom{4}{1}\binom{6}{2} = 330 - 140 - 150 + 60 = \boxed{100}.$$

RR 4 Part 2: Let $T = TNYWR$. Regular octagon $OLYMPIAD$ is inscribed in circle Q. Circle Q has area $T\pi$. The area of octagon $OLYMPIAD$ can be expressed as $a\sqrt{b}$, where a and b are positive integers and b is not divisible by the square of any prime. Find $a + b$.

Answer: $\boxed{202}$

Solution: Since the area of circle Q is $T\pi = 100\pi$, the radius of circle Q is 10. Let the center of circle Q be Q_1. Since $\angle OQ_1L = 45°$, from the Law of Cosines on OQ_1L, we have

$$OL^2 = 10^2 + 10^2 - 2 \cdot \left(\frac{\sqrt{2}}{2}\right)(10)(10) = 200 - 100\sqrt{2}.$$

Now, since the area of a regular octagon with side length s is $s^2(2 + 2\sqrt{2})$, we have

$$[OLYMPIAD] = (200 - 100\sqrt{2})(2 + 2\sqrt{2}) = 200\sqrt{2} \implies 200 + 2 = \boxed{202}.$$

RR 4 Part 3: Let $T = TNYWR$. Real numbers a, b, and c satisfy

$$a + b = -c,$$
$$a^3 - abc = 4, \text{ and}$$
$$b^3 - abc = T.$$

Find the value of $abc - c^3$.

Answer: $\boxed{206}$

Solution: Since $a + b = -c \implies a + b + c = 0$, we have

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - ac - bc) = 0.$$

So,

$$(a^3 - abc) + (b^3 - abc) - (abc - c^3) = 0 \implies abc - c^3 = 4 + T = 4 + 202 = \boxed{206}.$$

2024 SSMO Relay Round 5 Solutions

SMO Team

RR 5 Part 1: Let the super factorial $!(n)$ be defined on positive integers as $\prod_{i=1}^{n} i!$. Find the largest positive integer k such that there are exactly k positive integers n such that $!(n)$ has fewer than k trailing zeroes.

Answer: $\boxed{12}$

Solution: Let $f(n)$ be the number of trailing zeroes $!(n)$ has and $g(n)$ denote the number of trailing zeroes $n!$ has. Since $v_2(i!) \geq v_5(i!)$ for each i, the number of trailing zeroes of $!(n)$ is $\sum_{i=1}^{n} v_5(i!)$, so $f(n) = f(n-1) + g(n)$. Consider the following table.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14
$g(n)$	0	0	0	0	1	1	1	1	1	2	2	2	2	2
$f(n)$	0	0	0	0	1	2	3	4	5	7	9	11	13	15

So, exactly the 12 integers $n = 1, 2, \ldots, 12$ satisfy $f(n) < 12$, meaning $k = 12$ works. Since $g(n) \geq 2$ for $n \geq 10$, the table shows that $f(n) \geq 2n - 13$ for all n. So, if $f(n) < k$, then $n < \frac{k+13}{2}$, meaning at most $\frac{k+12}{2}$ positive integers n satisfy $f(n) < k$. Thus, $k \leq \frac{k+12}{2} \implies k \leq 12$, and the answer is $\boxed{12}$.

RR 5 Part 2: Let $T = TNYWR$. In the game of high and low, the computer chooses two integers uniformly at random without replacement from the set $\{1, 2, 3, \ldots, T\}$. The computer displays the first integer and asks the player if the second integer is higher or lower. Given that the player always plays optimally, the probability of guessing correctly can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{39}$

Solution: Note that if the computer displays $k \leq \frac{T}{2} = 6$, then the player guesses higher and is correct with probability $\frac{T-k}{T-1} = \frac{12-k}{11}$, and if $k \geq 7$, then the player guesses lower and is correct with probability $\frac{k-1}{11}$. Thus, the answer is

$$\frac{2(6 + \cdots + 11)}{11 \cdot 12} = \frac{17}{22} \implies 17 + 22 = \boxed{39}.$$

RR 5 Part 3: Let $T = TNYWR$. Let k be the largest prime factor of T. How many values of x satisfy

$$\sin(x^2) + \cos(x^2) = \sin^2 x + \cos^2 x \quad \text{and} \quad -k \leq x \leq k?$$

Answer: $\boxed{107}$

Solution: Note that $T = 39 = 3 \cdot 13$, so $k = 13$. Now, $\sin^2 x + \cos^2 x = 1$, so we are seeking to find the number of solutions to $\sin(x^2) + \cos(x^2) = 1$. The only solutions are when $x^2 = 2j\pi$ or $x^2 = \frac{\pi}{2} + 2j\pi$ for a nonnegative integer j. Now, for $-13 \leq x \leq 13$, we have $x^2 \leq 169$. Now, note that $52\pi + \frac{\pi}{2} \leq 169 < 54\pi$. So, there are 54 possible values for x^2, each giving two solutions, except $x^2 = 0$. In conclusion, the answer is $54 \cdot 2 - 1 = \boxed{107}$.